

Lab: Exception Handling with Integer Input Validation

Objective

Learn how to use a `try/except` block to handle exceptions and repeatedly prompt a user until a valid integer is entered.

Instructions

Write a Python program that:

1. Prompts the user to enter an integer.
 2. Uses a `try` block to convert the input to an integer.
 3. Uses an `except ValueError` block to catch invalid input.
 4. If the user enters a string (such as "abc") or a decimal value (such as "3.14"), display an error message and prompt them to try again.
 5. Continue prompting until a valid integer is entered.
 6. Display the valid integer entered by the user.
 7. Print "Thanks!" before the program ends.
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Sample Output #1

Plain Text

Enter an integer: hello

Invalid input. Please enter a whole number.

Enter an integer: 3.14

Invalid input. Please enter a whole number.

Enter an integer: 25

You entered a valid integer of 25.

Thanks!

Show more lines

Sample Output #2

Plain Text

Enter an integer: 100

You entered a valid integer of 100.

Thanks!
Show more lines

Starter Code

```
Python
# Write your code below

while True:
    try:
        # Prompt the user for an integer

        # Display the valid integer

        # Exit the loop

    except ValueError:
        # Display an error message

print("Thanks!")
``
```

Completed Solution

```
while True:
try:
    number = int(input("Enter an integer: "))
    print("You entered a valid integer of " + str(number) + ".")
    break
except ValueError:
    print("Invalid input. Please enter a whole number.")

print("Thanks!")
```

Discussion Questions

1. What exception is raised when a user enters non-numeric text?
2. Why is a `while True` loop used in this program?
3. What does the `break` statement do?
4. Why does entering `3.14` cause a `ValueError` when using `int()`?
5. How would you modify the program to accept decimal values as valid input?